MULTI-BRANCH BLOOD SUPPLY CHAIN OPTIMIZATION & EMERGENCY REDISTRIBUTION MESH

VitaSync: An IoT-Enabled Hospital Blood-Need Prioritization and Redistribution Support System

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Hackanight Technical Research Paper

Software-focused prototype proposal using synthetic or authorized inventory data

Research status: Proposed system and experimental plan

Abstract

Blood supply networks can experience local shortages even when stock exists elsewhere, while inventory approaching expiry may remain unused. VitaSync is a proposed decision-support system that connects hospital demand reporting with request prioritization, stock verification, route feasibility checks, and human-approved redistribution. The design combines an IoT-connected request workflow with a Random Forest demand-forecasting experiment, a priority queue, graph-based route calculations, and inventory-safe recommendation logic. In the completed software checks available for this revision, a chronological held-out test on 1,827 rows of synthetic surrogate data produced a Random Forest RMSE of 12.98, compared with 27.73 for a historical-usage baseline and 18.96 for a training-mean baseline. All 35 recorded validation checks passed for the tested notebook sections. These figures demonstrate that the prototype pipeline executes on synthetic data; they do not establish forecasting accuracy for real hospitals. The reinforcement-learning section was not run, and the proposed full MILP redistribution model has not been validated as a real-world deployment. VitaSync remains a logistics decision-support prototype and does not make patient-specific transfusion, compatibility, crossmatch, or blood-release decisions.

Keywords: blood supply chain, emergency redistribution, IoT communication, Random Forest, inventory verification, route feasibility, priority queue, MILP, synthetic data.

1.Introduction

Blood and blood components are essential for trauma care, surgery, obstetric emergencies, paediatric care, and treatment of blood disorders. A reliable service must coordinate collection, testing, component preparation, storage, inventory visibility, transport, and authorized release. Unlike ordinary goods, blood components are perishable and have component-specific storage and handling requirements. A stock entry at one facility is not automatically available or suitable for transfer to another facility.

VitaSync is designed as a coordination layer that turns hospital-reported requirements into a time-stamped, explainable priority queue and links requests to verified-stock and route-feasibility checks. It is intended to complement existing blood-bank systems rather than replace them. The updated paper distinguishes the proposed architecture from the parts that have actually been tested in a synthetic prototype.

2. Problem Statement

Inventory, demand, and transport capacity may be unevenly distributed across a regional network. One centre may have eligible stock while another reports an urgent need. A transfer may still be unsuitable if stock data are stale, units are reserved or unreleased, the source would fall below its minimum reserve, the route cannot meet the deadline, or transport conditions cannot be confirmed. The research question is how to coordinate requests and identify feasible redistribution options while reducing unmet demand and expiry exposure without bypassing authorized safety rules.

3. Aim, Objectives and Scope

3.1 Aim

To develop and evaluate a software prototype for connected hospital blood-need reporting, transparent request prioritization, short-horizon demand forecasting, route-aware recommendations, and inventory-safe redistribution support.

3.2 Objectives

- Receive and timestamp hospital demand requests through a connected interface or gateway.
- Forecast demand using a Random Forest model and compare it with simple baselines.
- Rank requests using a documented urgency policy and predictable tie-breaking.
- Represent facilities and routes as a graph and calculate travel-time paths.
- Exclude stale, expired, reserved, unreleased, unverified, or otherwise ineligible inventory from transfer recommendations.
- Preserve source-centre reserves and prevent duplicate reservations or negative inventory in simulated scenarios.
- Keep approval, release, compatibility, and clinical decisions under authorized blood-service personnel.

3.3 Scope and boundaries

The tested forecasting data are synthetic and cover 1 January 2020 to 31 December 2024. Demonstration facilities, demand requests, stock levels, urgency scores, and travel times are fictional. The current evidence does not prove live IoT hardware integration,

production routing, clinical compatibility, or real-world patient benefit. Reinforcement learning using PPO was not executed in the supplied proof of working.

4. Background and Related Work

Blood supply chain literature identifies uncertainty, perishability, shortage risk, outdating, and distribution as interdependent planning problems. Research surveys classify blood supply chain decisions across network design, inventory management, operational processes, and solution methods. In India, blood transfusion services have developed through safety programmes, regulation, and digital inventory initiatives. The Government of India's e-RaktKosh platform provides centralized blood-bank management and inventory-related services. VitaSync is framed as a complementary workflow for timestamped hospital demand intake, explainable urgency ranking, stock confirmation, and human-approved transfer proposals.

These sources establish the importance of coordinated inventory and distribution but do not validate VitaSync itself. Its contribution is a proposed integration of connected demand intake, forecasting, queue prioritization, route checks, and traceable approval. A systematic prior-art review would be required before making any claim that the approach is the first of its kind.

5. Proposed System Architecture

Layer	Function	Output
1. Demand intake	Authorized staff submit component category, quantity, urgency, required-by time, and facility.	Validated, time-stamped request
2. Connectivity / gateway	Moves requests and status updates between participating facilities and the application.	Connected event record; protocol must be documented after implementation
3. Forecasting	Random Forest estimates demand from calendar and operational features.	Forecast and evaluation metrics
4. Priority queue	Ranks active requests using defined urgency factors and deterministic tie-breaking.	Explainable ranked queue
5. Inventory and route checks	Checks stock status, reserves, quantities, route times, and deadline feasibility.	Feasible options or review status
6. Approval and audit	Source and receiving staff review; pickup, custody, delivery, and exceptions are recorded.	Approved or pending manifest and audit trail

IoT note: the current project describes an IoT-connected communication workflow, but the available evidence does not confirm a specific deployed microcontroller, sensor suite, radio protocol, or live vehicle-tracking system. GPS, temperature sensors, LoRa, and offline message queues should be described as proposed options unless implemented and tested. The ResQtech concept informs the design principles of layered communication, route awareness, resilience, authentication, and fail-safe handling; its traffic-signal control is not part of VitaSync.

6. Application and 3D Scenario Demonstration

To support the system-design description, three demonstration resources are provided below. The live application link opens the deployed VitaSync web interface; the application demonstration provides a visual walkthrough, while the 3D scenario video illustrates the proposed blood-request and transport process. These links are supplementary demonstration materials; they are not, by themselves, evidence of clinical validation or a live operational deployment.

6.1 Live Application: Open the VitaSync Web Application

6.2 Application Walkthrough: Open the VitaSync Application Demonstration

This resource is included as a visual walkthrough of the application. Readers can use it to follow the available screens and features. The exact functions shown should be described in the final submission only to the extent that they are visible and implemented in the demonstration.

6.3 3D Scenario Video: https://drive.google.com/file/d/1DKLGvxUZY_NdQLTfcQDsTFpkGQM9-rm0/view?usp=sharingThe 3

scenario provides a conceptual visualization of how a hospital request may lead to stock checking, source selection, route planning, transport, and delivery confirmation. It should be interpreted as a scenario illustration rather than a measured test of real-world delivery time or sensor performance.

7. Unique Blood-Transport Workflow

1. Request and alert: an authorized hospital manager submits the requirement, urgency, quantity, and deadline; the system records the timestamp.
2. Find suitable stock: the system checks confirmed, released, unreserved, unexpired inventory and protects the source centre's minimum reserve.
3. Choose a feasible route: candidate sources are assessed using estimated travel time, deadline, remaining shelf life, capacity, and validated transport requirements—not distance alone.
4. Approve and prepare: source and receiving staff review the proposed transfer and confirm a manifest before inventory is committed.
5. Track and reassess: journey status and, if validated sensors are installed, transport-condition evidence may be recorded. Delays or exceptions trigger review and possible replanning.
6. Confirm delivery: authorized staff confirm receipt and update inventory; custody, approvals, and exceptions remain auditable.

The key distinction is that VitaSync links need, stock eligibility, and route feasibility in one proposed workflow. It does not automatically send blood or autonomously authorize a transfer.

8. Data and Experimental Method

8.1 Forecasting dataset

The executed proof used a schema-matched synthetic surrogate dataset, not live Kaggle or hospital data. It contained 1,827 dated rows from 2020-01-01 through 2024-12-31. The target was PredictedBloodDemand. The eight input features were DayOfWeek, Month, Population, Events, HistoricalBloodUsage, HospitalAdmissions, BloodDonorsAvailable, and Temperature.

Partition	Rows	Use
Training	1,278	Fit model parameters
Validation	274	Compare settings / select configuration
Held-out test	275	Final chronological evaluation

The split was chronological to reduce the risk of training on future records when evaluating earlier demand. The selected Random Forest configuration used min_samples_leaf = 5 and max_features = 0.6. The reported test values are from the supplied machine-readable training proof.

8.2 Evaluation metrics

Mean Absolute Error (MAE) is the average absolute difference between predictions and target values. Root Mean Squared Error (RMSE) gives greater weight to larger errors. R^2 describes variance explained relative to a mean-prediction reference. Lower MAE and RMSE are preferable; higher R^2 is generally preferable. The synthetic target units are labelled as dataset demand units and must not be interpreted as verified clinical blood units.

9. Results

9.1 Held-out demand forecast

Method	Test MAE	Test RMSE	Test R^2
Random Forest	10.22	12.98	0.507
Historical-usage baseline	22.66	27.73	-1.251
Training-mean baseline	14.12	18.96	-0.052

On this synthetic held-out test, Random Forest had the lowest RMSE among the three compared methods. Its RMSE was 12.98, compared with 27.73 for the historical-usage baseline and 18.96 for the training-mean baseline. This is evidence of comparative performance on the generated dataset only; it is not evidence that the model will forecast real hospital demand with the same error.

9.2 Prototype validation checks

Area	Observed result
Data quality and split	Rows and dates valid; no duplicate dates; all eight features present; chronological split covers all rows.
Forecast	Metrics finite; Random Forest RMSE lower than both tested baselines; chart saved.

Route graph	Four nodes and five edges; graph connected; Depot-A to Site-C path via Site-B estimated at 28 minutes.
Priority queue	Heap order matched independent sorting; invalid urgency rejected; score rose with urgency; scores remained in [0,1].
Inventory recommender	Statuses valid; no site oversubscribed; ledger never negative; input inventory not mutated.
Deadline edge case	SIM-101 estimated at 28 minutes against a 12-minute deadline; status LATE_ROUTE_REVIEW; no stock reserved.
Local fulfillment	SIM-102 and SIM-103 were classified as local fulfillment review; zero-minute local route.
Artifacts	Saved model reloaded with identical predictions; feature schema matched; proof JSON, model, and chart existed.

The supplied proof records 35/35 checks passed for notebook sections 1–6. It also records an additional rerun with stand-ins for reinforcement-learning libraries, but that is not equivalent to training or validating a reinforcement-learning policy.

9.3 Illustrative route example

The synthetic request SIM-101 needed five units at Site-C with a 12-minute deadline. The selected simulated route from Depot-A via Site-B took 28 minutes, giving deadline slack of −16 minutes. The recommender correctly labelled the route for review and did not reserve stock. SIM-102 and SIM-103 were locally fulfilled in the toy scenario, with zero travel time. These are software edge-case demonstrations, not real deliveries.

10. Mathematical Optimization Formulation

The proposed full allocation model is a mixed-integer linear programming (MILP) formulation. Let $x_{ij}cg$ be the non-negative integer quantity proposed from source i to destination j for component/category (c,g), and $u_j cg$ the unmet demand. Let $S_i cg$ be verified eligible stock, $R_i cg$ the minimum source reserve, $D_j cg$ the destination demand, q_{ij} route capacity, and $F_{ij}cg$ a feasibility flag based on approved eligibility and transport checks.

A simplified objective is to minimize $Z = P_u \Sigma_j cg\ u_j cg + \Sigma_{ij} cg\ (C_{ij} + L_{ij} + E_i cg) x_{ij} cg$, where unmet demand receives a high penalty and transfer terms represent transport effort, delay risk, and expiry exposure. Constraints include demand accounting, source reserve protection, route and product eligibility, transport capacity, and integer quantities. Coefficients must be justified and validated for the experiment; they are not universal clinical constants.

Important implementation status: the available proof validates toy recommendation and inventory-safety checks, but it does not establish that the complete MILP model has been solved and benchmarked on real inventory. The full model and its parameters require further reproducible testing.

11. Discussion

The current results show that a reproducible software pipeline can be tested end-to-end on surrogate data, and that the forecast model outperformed two simple baselines under the synthetic test setup. The route and inventory checks also demonstrate useful guardrails: infeasible arrival times can be flagged, local fulfillment can be distinguished from transfer, and simulated stock is not reserved for a late route. These results support the feasibility of further prototype development, not operational effectiveness.

The largest limitation is data realism. Synthetic values may encode patterns that do not occur in actual hospitals. No inference should be made about national shortage rates, deaths prevented, blood wastage reduced, money saved, or clinical outcomes from this experiment. Forecast performance must be re-evaluated on authorized, representative data before any such claim.

12. Security, Safety, Ethics and Limitations

- Use synthetic or authorized de-identified data and avoid patient identifiers in dashboards.
- Treat reported inventory as unverified until confirmed by authorized staff.
- Apply only compatibility, product eligibility, release, and transport rules approved by qualified blood-service professionals.
- Require source and receiving-centre approval and retain an audit trail.
- Do not commit stock if the route misses the deadline or transport conditions cannot be validated.

- Protect communication with authentication, role-based access, replay/error handling, and secure update practices.
- Do not claim live GPS, cold-chain sensors, LoRa, offline fallback, or hardware fail-safe behaviour unless implemented and tested.
- The system is not clinical software and must not perform crossmatching, patient-specific transfusion decisions, or autonomous blood release.

The PPO/reinforcement-learning section was not run because its dependencies were unavailable in the recorded environment. The evidence must not be presented as a trained RL policy or as a completed comparison with MILP.

13. Conclusion and Future Work

VitaSync proposes a connected, explainable workflow for hospital blood-need prioritization and emergency redistribution. The updated experimental evidence includes a Random Forest forecast test on synthetic data, graph-based route checks, priority queue validation, and toy inventory-safety checks. The Random Forest test RMSE was 12.98, lower than the historical-usage and training-mean baselines in the same synthetic experiment. Thirty-five recorded checks passed for the executed notebook sections.

The results remain preliminary because the data are synthetic, the PPO experiment is pending, and real hospital integration has not been demonstrated. Future work should run the notebook with a verified real or authorized de-identified dataset, document data provenance, train and evaluate PPO only if appropriate, implement and benchmark the full MILP allocation model, test larger networks and sensitivity scenarios, validate IoT devices and communication security, and conduct a supervised shadow-mode pilot with licensed blood centres. All clinical and release decisions must remain with authorized professionals.

References

[1] R. N. Makroo et al., “A review of legal, regulatory, and policy aspects of blood transfusion services in India: Issues, challenges, and opportunities,” Asian Journal of Transfusion Science, 2021. Available: https://pmc.ncbi.nlm.nih.gov/articles/PMC8628249/

[2] Ministry of Health and Family Welfare, Government of India, “About e-RaktKosh: A Centralized Blood Bank Management System.” https://eraktkosh.mohfw.gov.in/BLDAHIMS/bloodbank/about.cnt

[3] J. Beliën and H. Forcé, “Supply chain management of blood products: A literature review,” European Journal of Operational Research, vol. 217, no. 1, pp. 1–16, 2012. doi: 10.1016/j.ejor.2011.05.026.

[4] A. F. Osorio et al., “Survey on blood supply chain management: Models and methods,” Computers & Operations Research, vol. 112, article 104756, 2019. doi: 10.1016/j.cor.2019.07.014.

[5] World Health Organization, “Strategic framework for blood safety and availability,” 2020. https://www.who.int/publications/i/item/strategic-framework-for-blood-safety-and-availability

[6] Directorate General of Health Services, Ministry of Health and Family Welfare, Government of India, “Blood Transfusion Services.”

[7]VitaSync team. Research paper and supporting research materials. Google Drive folder. Available: https://drive.google.com/drive/folders/1XtwMaGu3fFs3acIj6JYNqvLEWdMCQygo

[8] VitaSync prototype proof of working and training record supplied with the project: synthetic surrogate run, environment details, model metrics, and 35 validation checks, dated 2026.

[9] VitaSync team. Supporting proof materials. Google Drive folder. Available: https://drive.google.com/drive/folders/1ZfqBFbUnjgFyOAQL2ZAqHSyfSjqI2Lwx

Appendix A. Reproducibility and evidence boundary

Recorded environment: Python 3.12.3, NumPy 2.4.4, pandas 3.0.2, scikit-learn 1.8.0, NetworkX 3.6.1, Matplotlib 3.10.8, and joblib 1.5.3. The data mode was SURROGATE. Dataset size was 1,827 rows, split into 1,278 training, 274 validation, and 275 test rows. The trained model file SHA-256 recorded in the proof is e35ee9947bd5698afcb67ef3658651c42ebc7a214767ef93185d088792bde33c.

Before external submission, the team should verify reference metadata, add the actual IoT hardware/protocol details only if built, replace any remaining proposal claims with measured results, and clearly retain the synthetic-data disclaimer.